

CKA Exam Objectives → Course Map

Maps every objective in the official [CKA Exam Curriculum v1.35](#) (Cloud Native Computing Foundation) to where it's covered in this course. Objective wording is verbatim from the CNCF curriculum.

Students preparing for the [Certified Kubernetes Administrator](#) exam can use this to find the lab or slide section for a given topic, and to see at a glance what falls outside this 4-day course and needs additional self-study.

Domain Weights

Domain	Weight
Troubleshooting	30%
Cluster Architecture, Installation and Configuration	25%
Services and Networking	20%
Workloads and Scheduling	15%
Storage	10%

25% — Cluster Architecture, Installation and Configuration

Objective	Covered in
Manage role based access control (RBAC)	Day 4 slides — RBAC , Lab: RBAC and Access Control
Prepare underlying infrastructure for installing a Kubernetes cluster	Lab: Provisioning the Cluster

Objective	Covered in
Create and manage Kubernetes clusters using kubeadm	Lab: kubeadm Cluster Initialization
Manage the lifecycle of Kubernetes clusters	Lab: Cluster Upgrade , Lab: etcd Backup and Restore , Lab: Multi-Node Cluster Operations
Implement and configure a highly-available control plane	Day 4 slides — HA control plane concepts (join --control-plane, stacked etcd)
Use Helm and Kustomize to install cluster components	Helm — Lab: Monitoring with Prometheus and Grafana (kube-prometheus-stack). Kustomize is not covered in this course — self-study needed
Understand extension interfaces (CNI, CSI, CRI, etc.)	Lab: CNI Plugin Installation (Calico); containerd (CRI) in Lab: kubeadm Cluster Initialization ; CSI/StorageClass in Day 3 — storageclass-linode.yaml
Understand CRDs, install and configure operators	Gateway API CRDs in Day 3 slides; Prometheus Operator + PrometheusRule CRD in Lab: Monitoring with Prometheus and Grafana

20% — Services and Networking

Objective	Covered in
Understand connectivity between Pods	Lab: CNI Plugin Installation , Lab: Services and DNS
Define and enforce Network Policies	Day 3 manifests — netpol-allow-frontend.yaml , netpol-deny-all-ingress.yaml , netpol-deny-all-egress.yaml
Use ClusterIP, NodePort, LoadBalancer service types and endpoints	Lab: Services and DNS , Lab: NodePort and LoadBalancer , Lab: Expose the Application
Use the Gateway API to manage Ingress traffic	Day 3 slides — Gateway API, gatewayclass-nginx.yaml , gateway-prod.yaml , httproute-web.yaml
Know how to use Ingress controllers and Ingress resources	Lab: Ingress

Objective	Covered in
Understand and use CoreDNS	Lab: Services and DNS ; DNS troubleshooting in Lab: Namespaces in Practice

15% — Workloads and Scheduling

Objective	Covered in
Understand application deployments and how to perform rolling update and rollbacks	Lab: Rolling Updates and Rollbacks
Use ConfigMaps and Secrets to configure applications	Lab: ConfigMaps and Secrets
Configure workload autoscaling	Day 2 slides — Horizontal Pod Autoscaler, hpa-nginx.yaml
Understand the primitives used to create robust, self-healing, application deployments	Lab: Deploy a Multi-Replica Application , Lab: Health Checks , Lab: DaemonSet and CronJob
Configure Pod admission and scheduling (limits, node affinity, etc.)	Lab: Scheduling (taints/tolerations, node affinity); resource limits in Lab: Deploy a Multi-Replica Application ; Pod Security Standards / admission in Day 4 slides

10% — Storage

Objective	Covered in
Implement storage classes and dynamic volume provisioning	Lab: Persistent Storage , storageclass-linode.yaml
Configure volume types, access modes and reclaim policies	Lab: Persistent Storage , pv-manual.yaml
Manage persistent volumes and persistent volume claims	Lab: Persistent Storage , pvc-manual.yaml , pvc-app-storage.yaml

30% — Troubleshooting

Objective	Covered in
Troubleshoot clusters and nodes	Lab: Cluster Validation , Lab: Multi-Node Cluster Operations
Troubleshoot cluster components	Lab: Exploring Core Components (etcd, static pods, certificates, kubelet)
Monitor cluster and application resource usage	kubectl top, Lab: Monitoring with Prometheus and Grafana
Manage and evaluate container output streams	Lab: Troubleshoot a Broken Pod , Lab: Onboard a Pod (kubectl logs, exec)
Troubleshoot services and networking	Lab: Namespaces in Practice (cross-namespace DNS), Day 4 slides — systematic troubleshooting for pods/nodes/networking

Not Covered in This Course

A few CKA objectives fall outside this course's scope and need dedicated self-study:

- **Kustomize** — the course covers Helm for installing cluster components, but not Kustomize overlays/patches.
- **Writing/authoring CRDs and operators** — the course installs and uses operators (Prometheus Operator, Gateway API CRDs) but doesn't cover authoring a CRD or operator from scratch.

See the [Kubernetes from Scratch course](#) on KubeSkills for a deeper dive into these and other advanced topics (cluster internals, controllers, operators).